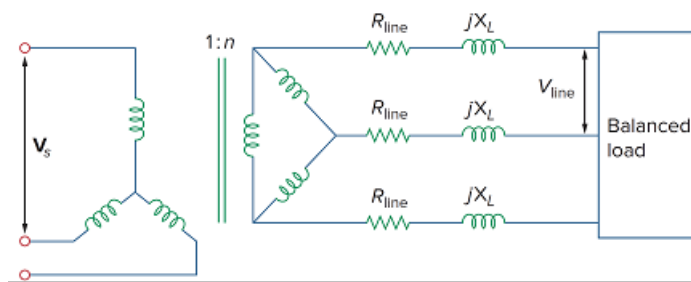


Problem

Using Fig. 13.138, design a problem to help other students better understand a Y-Δ, three-phase transformer and how they work.

Figure 13.138



Step-by-step solution

Step 1 of 5

Using Fig. 13.138, design a problem to help other students better understand a wye-delta, three-phase transformer and how they work.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

A Y-Δ three-phase transformer is connected to a 60-kVA load with 0.85 power factor (leading) through a feeder whose impedance is $0.05 + j0.1\Omega$ per phase, as shown in Fig. 13.137 below. Find the magnitude of:

(a) the line current at the load,

Step 2 of 5

(b) the line voltage at the secondary side of the transformer,

Step 3 of 5

(c) the line current at the primary side of the transformer.

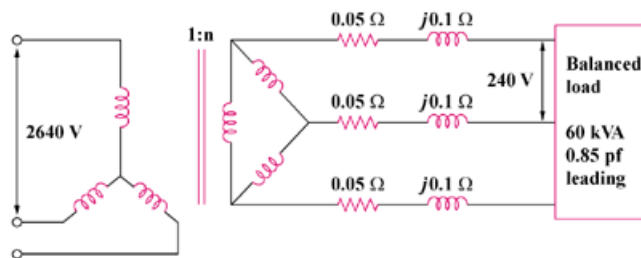


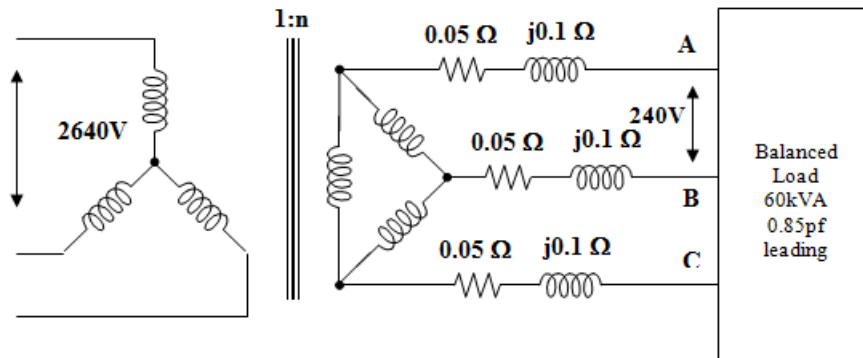
Figure 13.137

Solution

(a) At the load, $V_L = 240 \text{ V} = V_{AB}$

$$V_{AN} = V_L / \sqrt{3} = 138.56 \text{ V}$$

$$\text{Since } S = \sqrt{3} V_L I_L \text{ then } I_L = 60,000 / (240 \sqrt{3}) = 144.34 \text{ A}$$



Step 4 of 5

(b) Let $V_{AN} = |V_{AN}| \angle 0^\circ = 138.56 \angle 0^\circ$

$$\cos \theta = \text{pf} = 0.85 \text{ or } \theta = 31.79^\circ$$

$$I_{AA'} = I_L \angle \theta = 144.34 \angle 31.79^\circ$$

$$V_{A'N'} = Z I_{AA'} + V_{AN}$$

$$= 138.56 \angle 0^\circ + (0.05 + j0.1)(144.34 \angle 31.79^\circ)$$

$$= 138.03 \angle 6.69^\circ$$

$$V_{Ls} = V_{A'N'} \sqrt{3} = 138.03 \sqrt{3} = 239.1 \text{ V}$$

Step 5 of 5

(c) For Y-Δ connections,

$$n = \sqrt{3} V_{Ls} / V_{ps} = \sqrt{3} \times 238.7 / 2640 = 0.1569$$

$$I_{Lp} = n I_L / \sqrt{3} = 0.1569 \times 144.34 / \sqrt{3} = 13.05 \text{ A}$$